

Nikhil Bathula

Senior Software Engineer | Backend, AI & Real-Time Systems

Hyderabad | Work from office / Open to Remote / Hybrid

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Professional Summary

Senior Software Engineer with **3+ years of experience** designing and delivering **scalable backend platforms, enterprise applications, and AI-driven real-time systems**. Strong background in **Java Full Stack development**, modern backend frameworks, and **cloud-native DevOps automation**. Proven hands-on experience building **LLM-powered semantic search platforms, resume parsing systems, and real-time voice AI assistants** using WebSockets and streaming pipelines. Known for end-to-end ownership, performance optimization, and translating complex requirements into reliable production systems.

Technical Skills

Programming Languages

- Java, JavaScript, TypeScript, Python
- Dart (Flutter), Kotlin (Android)
- C# (.NET)

Frontend & Mobile

- React, React Native
- Flutter (cross-platform mobile)
- HTML5, CSS3, Tailwind CSS

Backend & APIs

- Spring Boot (Java)
- Node.js (Express)
- FastAPI, Flask (async, WebSockets)
- ASP.NET Core
- RESTful APIs, GraphQL

AI / LLM & Search Systems

- OpenAI (GPT-4o, Realtime API, Whisper)
- Google Gemini (1.5 Flash, 2.0 Flash, Live API)
- Anthropic Claude
- Prompt engineering & structured JSON extraction
- Retrieval-Augmented Generation (RAG)
- Resume parsing & semantic search

Databases & Vector Search

- PostgreSQL, MySQL, SQLite
- Qdrant (self-hosted vector DB)
- Pinecone, Chroma
- Apache Solr

Voice, Streaming & Realtime

- Twilio Voice SDK (Web, Android, React Native)
- WebSocket-based audio streaming
- Real-time STT → LLM → TTS pipelines
- Low-latency response optimization

DevOps & Infrastructure

- Docker, Docker Compose
 - Nginx, Jetty
 - CI/CD pipelines (GitHub Actions)
 - Linux server administration (Hetzner)
 - GPU-backed inference servers
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Professional Experience

Software Engineer / Senior Software Engineer

D3E Studio India Pvt Ltd | August 2022 – Present

- Designed and delivered **enterprise-grade backend services** using **Java (Spring Boot)** and **Node.js**, supporting web and mobile platforms.
 - Migrated a **PDF generation engine** from a Dart-based library to a **server-side Java module** within 4 months, improving reliability and maintainability.
 - Built and optimized **cross-platform mobile applications** released on **Android and iOS** stores.
 - Designed and automated **cloud deployments** using **Docker, shell scripting, and Nginx**, enabling repeatable and zero-downtime releases.
 - Owned **CI/CD pipeline design and automation**, significantly reducing deployment effort and release errors.
 - Implemented secure **REST APIs**, authentication layers, and **real-time WebSocket communication**.
 - Integrated **Twilio Voice** for live call streaming and foreground call handling across **Android and React Native** applications.
 - Designed AI-driven **resume parsing and semantic retrieval systems** using **LLMs and vector databases**, enabling structured candidate search and ranking.
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Key Projects & Systems

AI Resume Parser & Semantic Search Platform

- Designed an AI pipeline to extract **structured metadata** (skills, experience, projects, durations) from **PDF and DOCX resumes**.
- Implemented **LLM-based entity extraction** and **multi-step prompting** for higher accuracy.
- Indexed resumes using **Qdrant and Pinecone** for semantic similarity search.
- Enabled recruiter-focused search, ranking, and filtering workflows.

Real-Time Voice AI Assistant

- Built a **real-time AI voice assistant** using **Twilio Voice WebSockets**.
- Implemented streaming pipelines: **Audio → STT (Whisper / Gemini) → LLM → TTS**.
- Optimized system for **sub-second AI response latency**.
- Used for automated data collection and conversational workflows.

Low-Code App Builder Platform

- Engineered reusable **code-generation modules** enabling one-click builds for **Android, iOS, and Web**.
- Built abstraction layers for server-client communication and dynamic widgets.
- Improved developer productivity through automated multi-platform deployment support.

Enterprise HR Management System

- Owned development of core HR modules including **employee lifecycle, roles, access control, attendance, and workflows**.
- Integrated backend APIs to unify HR data and streamline enterprise processes.

Enterprise Service & Hostel Management Applications

- Delivered scalable business modules for **service tracking, request handling, and task workflows**.
- Implemented offline caching, secure APIs, and role-based access control.

Education

B.Tech – Computer Science Engineering

Sri Chaitanya Technical Campus — 2025

Diploma – Electronics & Communication Engineering

Government Model Residential Polytechnic, Gajwel — 2022

Strengths

- Strong system design and debugging skills
- Applied AI integration (production-focused)
- Real-time and low-latency system development
- End-to-end ownership and delivery mindset